

PHD POSITION IN OPTIMIZATION UNDER UNCERTAINTY AT ROCHESTER INSTITUTE OF TECHNOLOGY

Position Summary:

We invite applications for a funded PhD position to conduct research in the field of **Optimization under Uncertainty** with Applications to Supply Chain Management and Healthcare Operations. The candidate will work under the supervision of [Dr. Aliaa Alnaggar](#) at the Department of Industrial and Systems Engineering, Rochester Institute of Technology (RIT). The successful applicant will lead research activities related to the development and implementation of optimization models. They will also receive regular training and mentorship, and will be expected to disseminate their research findings by publishing their research in academic journals.

Rochester Institute of Technology, located in Rochester, New York, is a top-tier research university internationally recognized for its leadership in science, technology, engineering, and experiential education. RIT's Kate Gleason College of Engineering provides state-of-the-art research facilities and a collaborative environment that fosters cutting-edge research, interdisciplinary innovation, and industry application.

Desired Qualifications:

1. MS or BS (earned or to be granted) in Industrial Engineering, Management Science, Operations Research, Applied Mathematics, Operations Management, Information Systems, Computer Science, or related fields.
2. Strong background in operations research and optimization. Exposure to Machine Learning techniques is an asset.
3. Strong communication skills (both written and spoken) and self-motivation.
4. Good programming skills with knowledge in Python, R, Java, or C/C++. Exposure to optimization software such as Gurobi and CPLEX is an asset.
5. Participation in an NSF REU (Research Experiences for Undergraduates) or application to/award of the NSF GRFP (Graduate Research Fellowship Program) is strongly encouraged and considered an asset.

This position is for a four-year PhD duration, with an anticipated start date of Fall 2027. Applicants already in the US may be considered for a Spring 2027 start date. Interested candidates should send their application materials in an email with the subject line "Prospective PhD Student" to

Dr. Aliaa Alnaggar

Email: amaeie@rit.edu

Please include your CV, a cover letter indicating why you would be appropriate for the position, contact information for references, and sample research papers or preprints (if applicable). Applications will be reviewed on a rolling basis.

Department of Industrial and Systems Engineering

Rochester Institute of Technology, Rochester, NY

<https://www.rit.edu/engineering/departments-industrial-and-systems-engineering>